

Binary Choice
Models

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Numerical
Methods

Theoretical
Concepts and
Estimation

Linear Probability
Model

Probit and Logit

Marginal
Effects

Average Marginal
Effects

Predicted
Probabilities

Additional
Example

Binary Choice Models

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Topics covered

- Linear probability model (largely irrelevant)
- Logit and probit models
- Marginal and predicted probabilities

Binary choice models and possible research questions

- Did you vote during the last election?
- Does an individual get arrested after being released from prison?
- Does an individual participate in the labor market?

Dependent variable y takes on one of two values: 0 or 1

Binary Choice as the Basis for Qualitative Choice

Conditional probability

$$P(Y = 1|X)$$

Binary choice part of broader class of so-called qualitative choice models

- Discrete dependent variable
- Nonlinear probability mapping, i.e., $P(Y = 1|X) = F(X \cdot \beta)$, for which linear (in coefficients) function $X \cdot \beta$ is transformed by nonlinear function $F(\cdot)$ to ensure predicted probabilities being between 0 and 1

Extensions in upcoming lectures

- Ordered outcomes
- Multinomial outcomes

Technical note: Notation of $X \cdot \beta$ instead of $\beta \cdot X$ adopted for mathematical reasons

Example

Consider the following equations

$$y = x^2$$

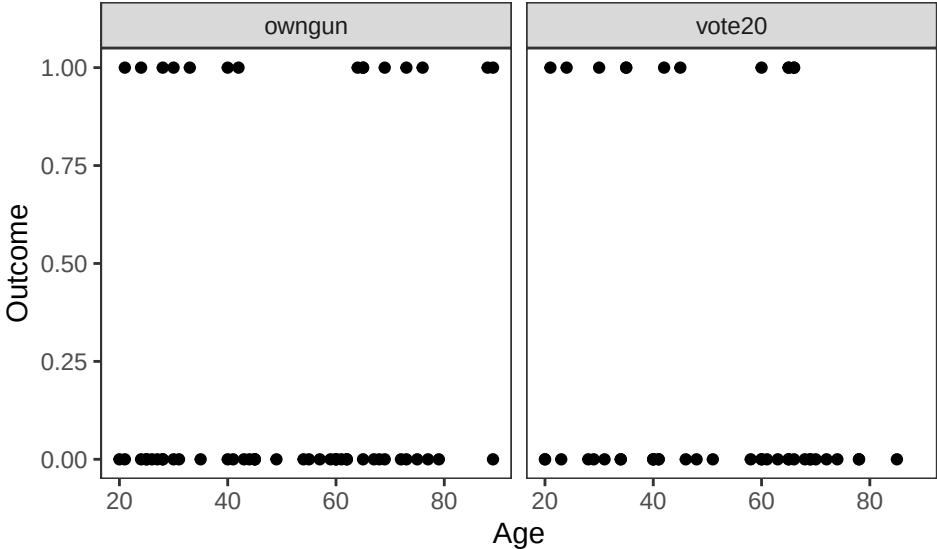
$$y = x^2 + \sqrt{x}$$

Questions

- What is the value of y if $x = 4$ or $x = 9$?
- What is the value of x if $y = 81$ or $x = 14$?

Need for numerical methods to determine the answer to the last question more generally for the second equation

Sample Binary Choice Data from the GSS



Linear Probability Model (LPM)

Most rudimentary model for binary choice

- Use of linear regression model, i.e., $y_i = \beta_0 + \beta_1 \cdot x_i + \epsilon$

Problems

- Possibility of $E(y_i|x_i) > 1$ or $E(y_i|x_i) < 0$
- Error terms are neither normally distributed nor homoscedastic
- Misspecification of functional form for probabilities

Alternative: Logit and Probit

- Calculation of the probability that the outcome equals 1 given exogenous variables, i.e., $Pr(y = 1|x)$

Latent Variable and Distributional Assumption

Unobserved latent variable Y^* representing net benefit or utility

$$Y^* = X \cdot \beta + \epsilon$$

Observed outcome Y

$$Y = \begin{cases} 1 & \text{if } Y^* > 0 \\ 0 & \text{if } Y^* \leq 0 \end{cases}$$

Distributional assumption about the error term

- Logit: ϵ follows logistic distribution
- Probit: $\epsilon \sim N(0, 1)$

Implied probability with $F(\cdot)$ determined by assumed distribution

$$P(Y = 1|X) = P(\epsilon > -X \cdot \beta) = F(X \cdot \beta)$$

Common Setup for Logit and Probit Models

General assumption about some function $G(\cdot)$ for all values of z

$$0 \leq G(z) \leq 1$$

Let

$$z = \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$$

Then, we have

$$P(y = 1|x) = G(\beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k)$$

Logit Model

Remember the Bernoulli distribution from statistics

$$P(y = 1) = p$$

$$P(y = 0) = 1 - p$$

with $E(y) = p$. Use of the cumulative logistic distribution function leading to the following for the logit model

$$Pr(y = 1) = G(z) = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}}$$

where $z = \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$

Probit Model

Instead of using the cumulative logistic distribution, the probit model uses the cumulative normal distribution

$$G(z) = \Phi(z)$$

Both models lead to similar results (not similar coefficients) and are solved using so-called maximum likelihood estimation (MLE)

Example using organic

Data description

- *income* of the respondent in \$ 1,000
- *buying* of organic food: yes (1) or no (0)

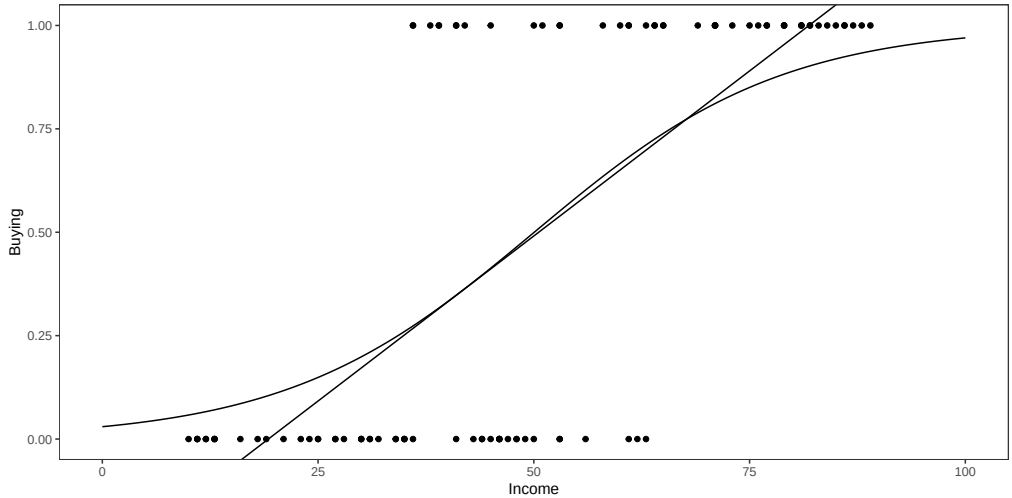
Results of interest for the binary choice model (for other models as well)

- Coefficient estimates
- Marginal effects
- Predicted probabilities

Notes

- Estimation through Maximum Likelihood
- Difficulty interpreting the values of coefficient

Comparison LPM vs. Logit



Estimation with R

Two possibilities for estimating coefficients in R

- Built-in R command `glm()`
- Function `logitmfx` or `probitmfx` from package `mfx`

Estimates using the `glm()`

```
bhatglm = glm(buying~income,family=binomial(link="logit"),  
              data=organic)
```

Estimates using the `logitmfx`

```
bhatmfx = logitmfx(buying~income,data=organic)  
summary(bhatmfx$fit) # Difference compared to previous use
```

Identical coefficient estimates using either `glm` or `mfx` package

Base Results

##	
##	=====
##	Dependent variable:
##	-----
##	buying
##	-----
##	income 0.117***
##	(0.022)
##	Constant -5.876***
##	(1.138)
##	-----
##	Observations 100
##	Log Likelihood -35.466
##	Akaike Inf. Crit. 74.931
##	=====
##	Note: *p<0.1; **p<0.05; ***p<0.01

Notes on Coefficient Estimates

Estimating probit instead of logit model

- Replace `family=binomial(link="logit")` with `family=binomial(link="probit")`
- Replace `logitmfx` with `probitmfx`

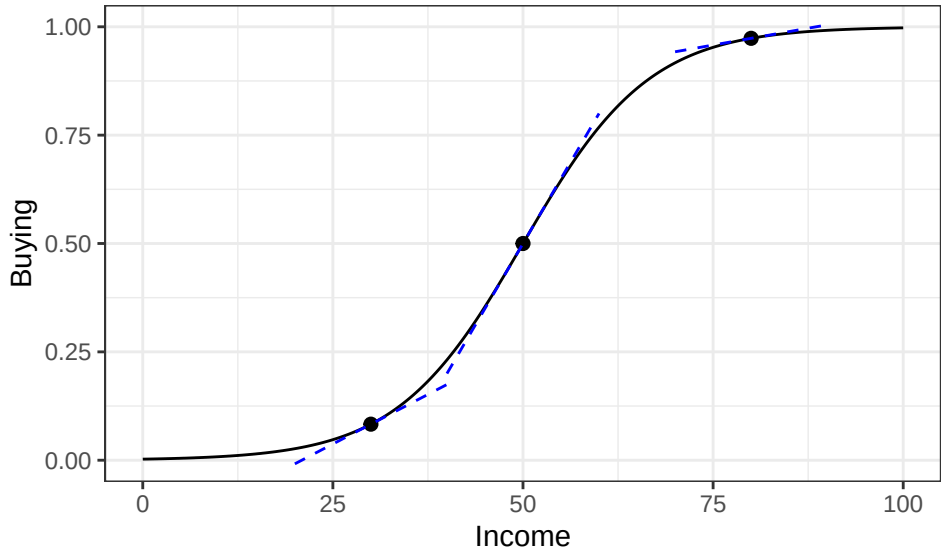
Different coefficient values between logit and probit

- Identical signs and usually very similar in terms of statistical significance

Interpretation

- Sign indicates only direction of probability change, e.g., positive coefficient for income leads to a higher (lower) probability of purchasing organic given a higher (lower) income

Difficulty of Interpreting Marginal Effects



Non-constant marginal effects

- Different change in probability depending on the values of the independent variables

Calculation of the marginal effect at the mean of the independent variables

- Use of output from [mfx](#)

Difference between marginal effects at a specific point and average marginal effect (AME)

Theoretical Aspects

Marginal effect at a specific point

$$\frac{\partial P(Y = 1|X)}{\partial X_k} = f(X \cdot \beta) \cdot \beta_k$$

Average Marginal Effect (AME) for variable k

$$ME_i = f(X_i \cdot \beta) \cdot \beta_k$$

Average across sample

$$AME = \frac{1}{N} \sum_{i=1}^N ME_i$$

Use of output from either `logitmfx` or `probitmfx`

```
bhatmfx$mfxest
```

```
##              dF/dx    Std. Err.          z      P>|z|  
## income 0.02919553 0.005634262 5.181785 2.197728e-07
```

Use of output from `glm()`

```
margins(bhatglm, at=data.frame(income=mean(organic$income)))
```

```
##   at(income) income  
##         49.31 0.0292
```

Average Marginal Effects

```
margins(bhatglm)
```

```
##      income
```

```
##    0.01305
```

Difference in value between marginal effect and average marginal effect

Fitted Values in a Binary Choice Model

Example

- What are the predicted probabilities of a person purchasing organic given their annual income (in \$ 1,000) of 30, 50, and 80?

Solution in R

```
datablock = data.frame(income=c(30,50,80))  
predict(bhatglm,newdata=datablock,type="response")
```

Food Purchases fpdata

Food purchases data

- `strawberries_org`: Frequency of strawberry purchases per month
- `tomatoes_org`: Frequency of strawberry purchases per month
- `age`: Age of the respondent
- `kidsunder12`: Presence of kids under the age of 12
- `rootsurban`: Urban (as opposed to rural) upbringing of respondent
- `education`: Education level
- `income`: Income

Data Preparation and Estimation

```
fpdata$strawberriesorg = ifelse(fpdata$strawberriesorg==0,0,1)
fpdata$tomatoesorg      = ifelse(fpdata$tomatoesorg==0,0,1)
```

```
bhats = glm(strawberriesorg~age+kidsunder12+rootsurban+
            education+income,
            family=binomial(link="logit"),
            data=fpdata)
bhatt = glm(tomatoesorg~age+kidsunder12+rootsurban+
            education+income,
            family=binomial(link="logit"),
            data=fpdata)
```


Additional Questions

For the strawberries and tomatoes regression, do the following

- Calculate the marginal effects of all independent variables
- Calculate the predicted probability for each observation